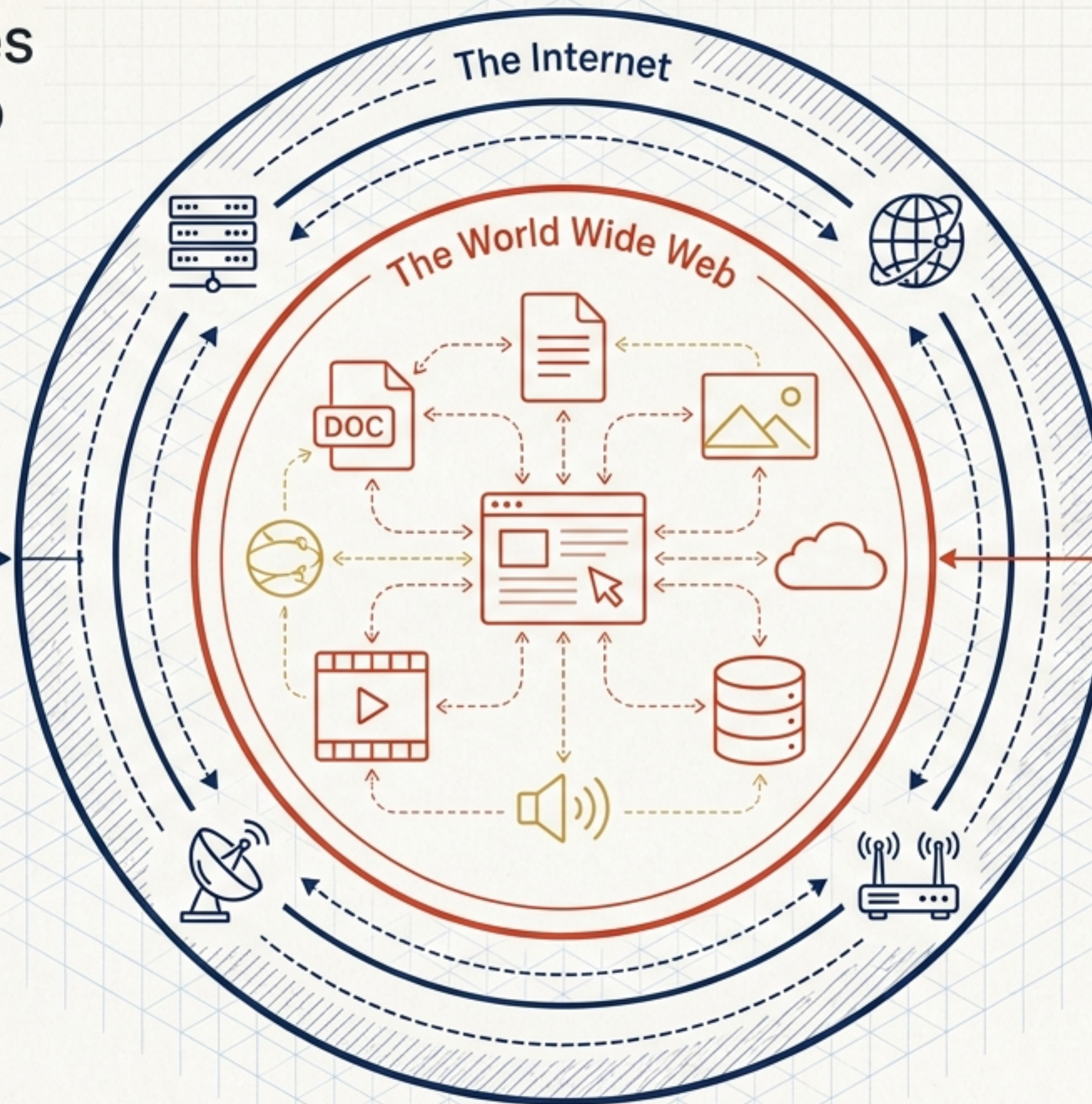


Mapping the Web Architecture

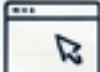
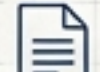
A structural guide to the protocols, systems, and languages that build the modern internet.

The Web Operates as a Layer on Top of the Internet

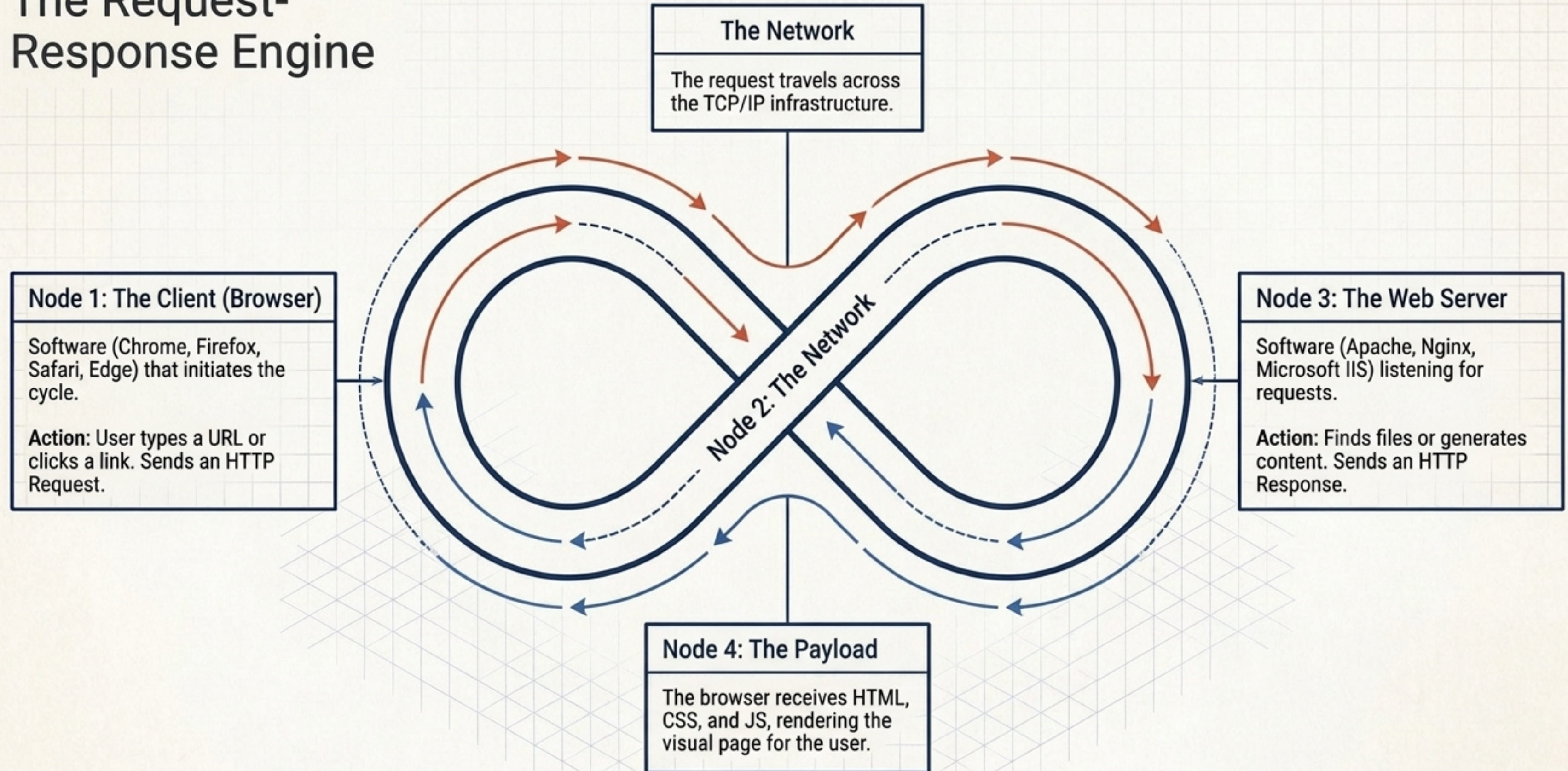
The Internet Container
A massive global network of interconnected devices. It provides the physical routing and infrastructure, communicating via TCP/IP (Transmission Control Protocol / Internet Protocol) for reliable data travel.



The Web Container
A system of interlinked hypertext documents and multimedia resources. It is just one of many services (alongside email and FTP) that run on the Internet's physical infrastructure.

Glossary Node	
	Website: A collection of related web pages under a common domain.
	Web Page: A single document, usually written in HTML.

The Request-Response Engine



Decoding the Uniform Resource Locator

`https://www.example.com:443/path/page.html?id=5#section`

Protocol

Defines the communication rules. "https" indicates a secure connection.

Domain / Host

The human-readable name of the target server.

Port

The specific server entry point (default 80 for http, 443 for https).

Path

The exact folder and file location on the server.

Query String

Extra data or parameters sent to the server for processing.

Fragment

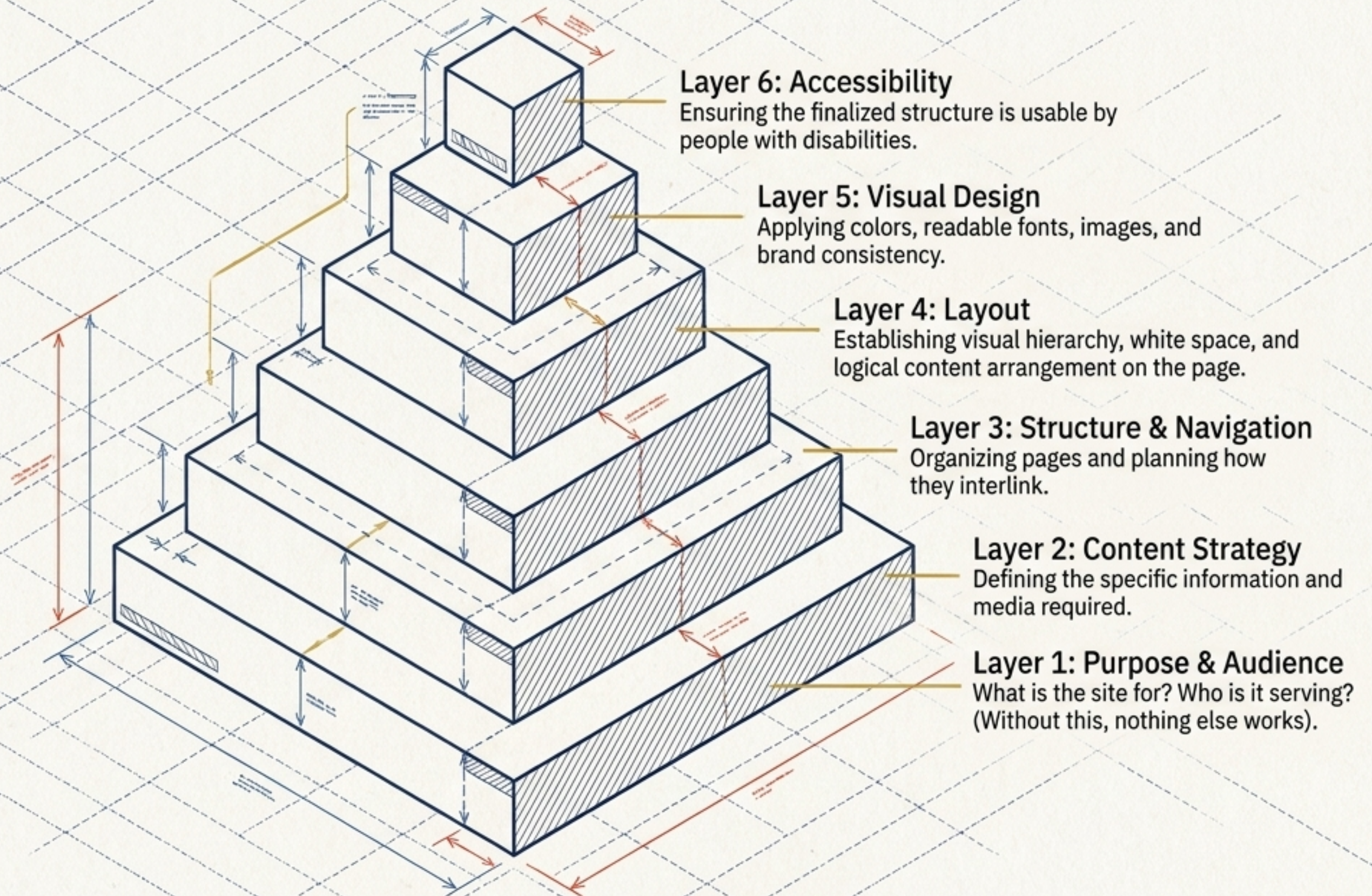
An anchor directing the browser to a specific location inside the returned page.

Architectural Paradigms: Static vs. Dynamic

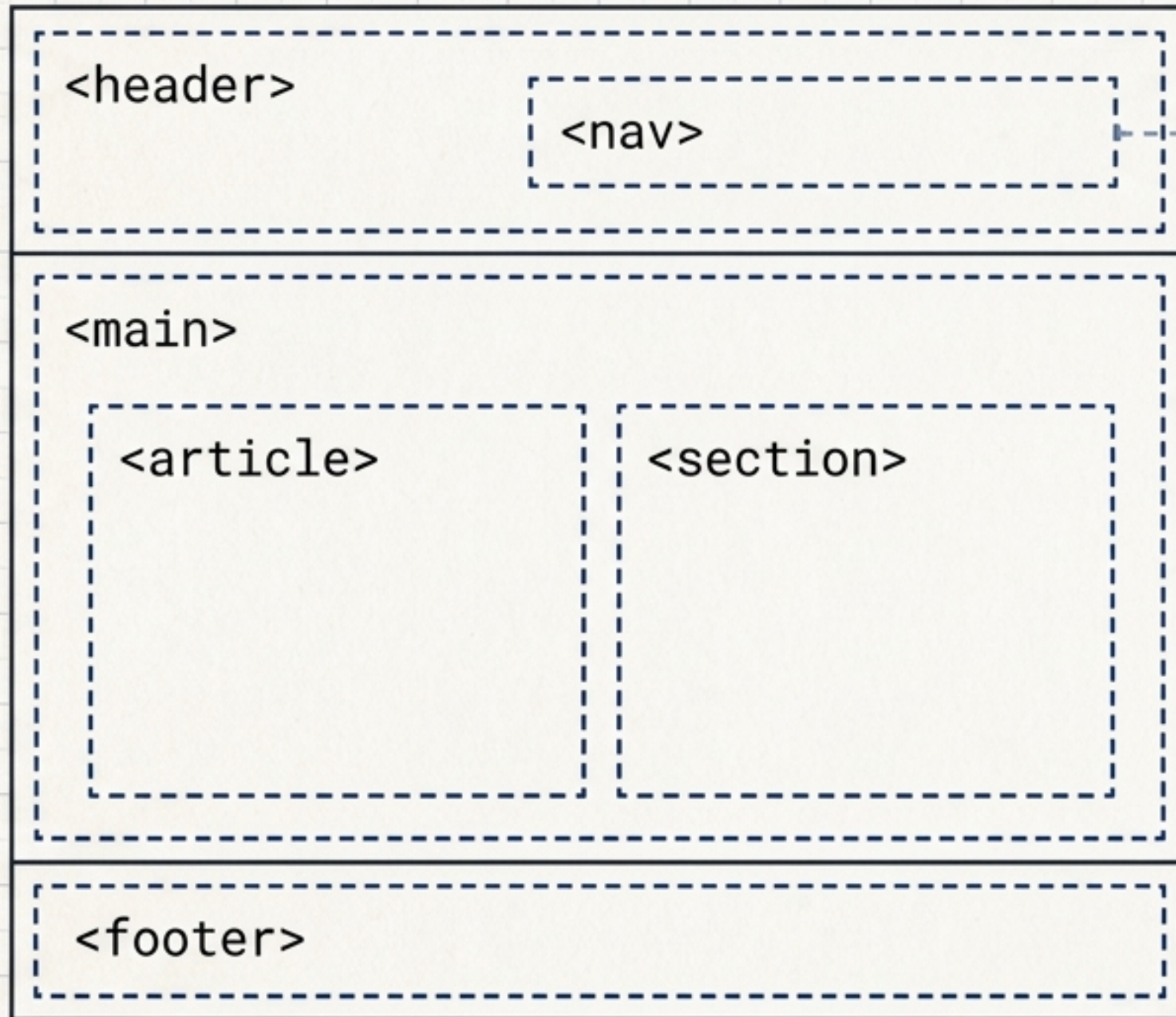
Matrix 1: The Architecture Matrix

	Static	Dynamic
Content Nature	Fixed in plain HTML files. The same for every visitor.	Generated on request. Can change per user, time, or context.
Under the Hood	Simple file storage.	Uses server-side programs (PHP, Python, Node.js) connected to Databases.
Maintenance & Updates	Requires manual developer edits to files.	Content managed via systems (like CMS) without touching code.
Deployment	Fast, simple, and cheap to host.	Requires heavier server processing and database hosting. (Used for shops, social networks).

The Foundation Stack: Planning Before Coding



The Structural Layer: HTML5 & Semantics



HTML (HyperText Markup Language) builds structure. HTML5 introduced **semantic elements** that describe the meaning of the content to machines, rather than just dictating appearance.

Core Build Components

- Text structure: `h1`, `p`
- Links: `a`
- Media: `img`, `video`
- Tables
- Forms

The Presentation Layer: Cascading Style Sheets (CSS)

Code Anatomy

Selector (Targets the HTML element).

```
h1 {  
  color: navy;  
  font-size: 24px;  
}
```

Declaration (Property: Value pairs defining the style).

Implementation

Three Ways to Deploy

1. **Inline:** Style attribute directly on a single HTML element.
2. **Internal:** Inside a **<style>** block in the HTML **<head>**.
3. **External:** A separate .css file linked via **<link>** tag (Best practice for production).

System Advantages

Enables consistent look across multiple pages, reduces HTML file sizes, and allows for responsive device adaptation.

Separation of Concerns: Structure vs. Presentation

Matrix 2: The Core Languages

HTML – The Skeleton

Domain: Content and Meaning.

Function: Defines what an element is (a paragraph, a heading, a list).

Machine Impact: Critical for Search Engine Optimization (SEO) as crawlers read the structure to index content.

Matrix 2: The Core Languages

CSS – The Paint

Domain: Layout and Aesthetics.

Function: Defines how an element looks (colors, spacing, positioning).

Maintenance Impact: One external stylesheet can command the design of thousands of HTML pages simultaneously.

Discoverability & Usability: SEO and Accessibility

Search Engine Optimisation (SEO) (Machine-focused)

Concept: Search engines (Google, Bing) use automated bots to crawl and index web pages.

Practice: SEO optimizes site structure and **semantic HTML** so search engines can accurately understand, categorize, and rank the content higher in search results.



Accessibility (Human-focused)

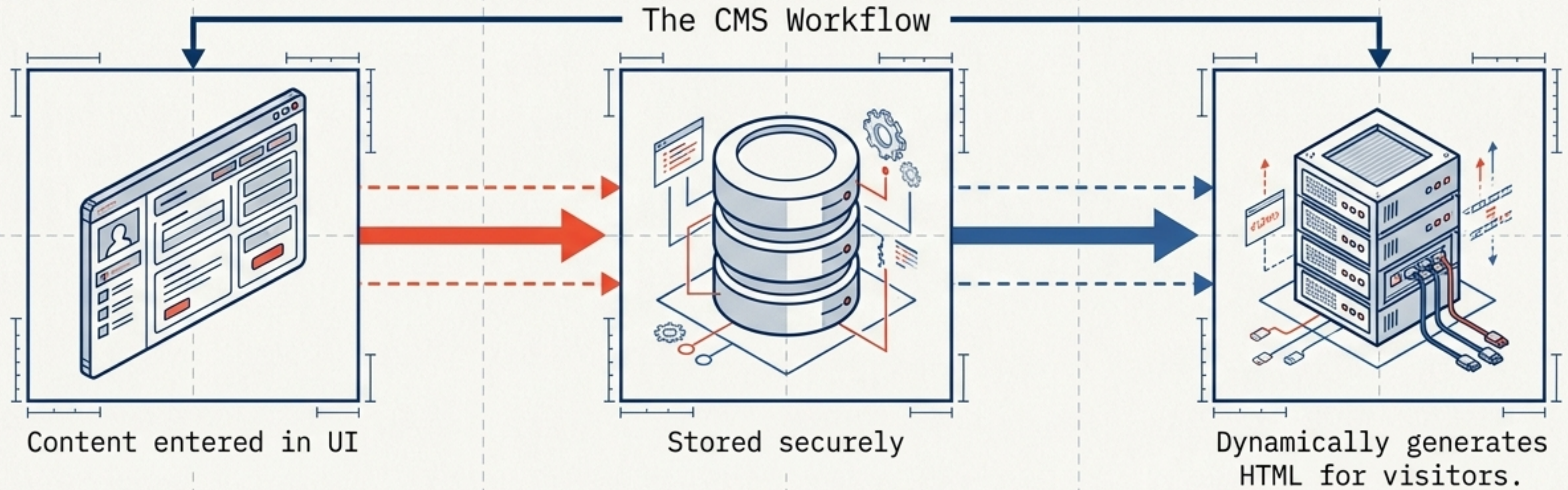
Concept: Designing interfaces usable by people with visual, hearing, motor, or cognitive disabilities.

Techniques: Requires proper heading hierarchies, descriptive **alt text** for images, sufficient color contrast, **keyboard-only navigation**, and precise **semantic HTML**.



Content Management Systems (CMS)

Software platforms that allow users to create, manage, and publish website content dynamically without writing code from scratch. Ideal for blogs and heavy business sites.



Industry Standards



WordPress

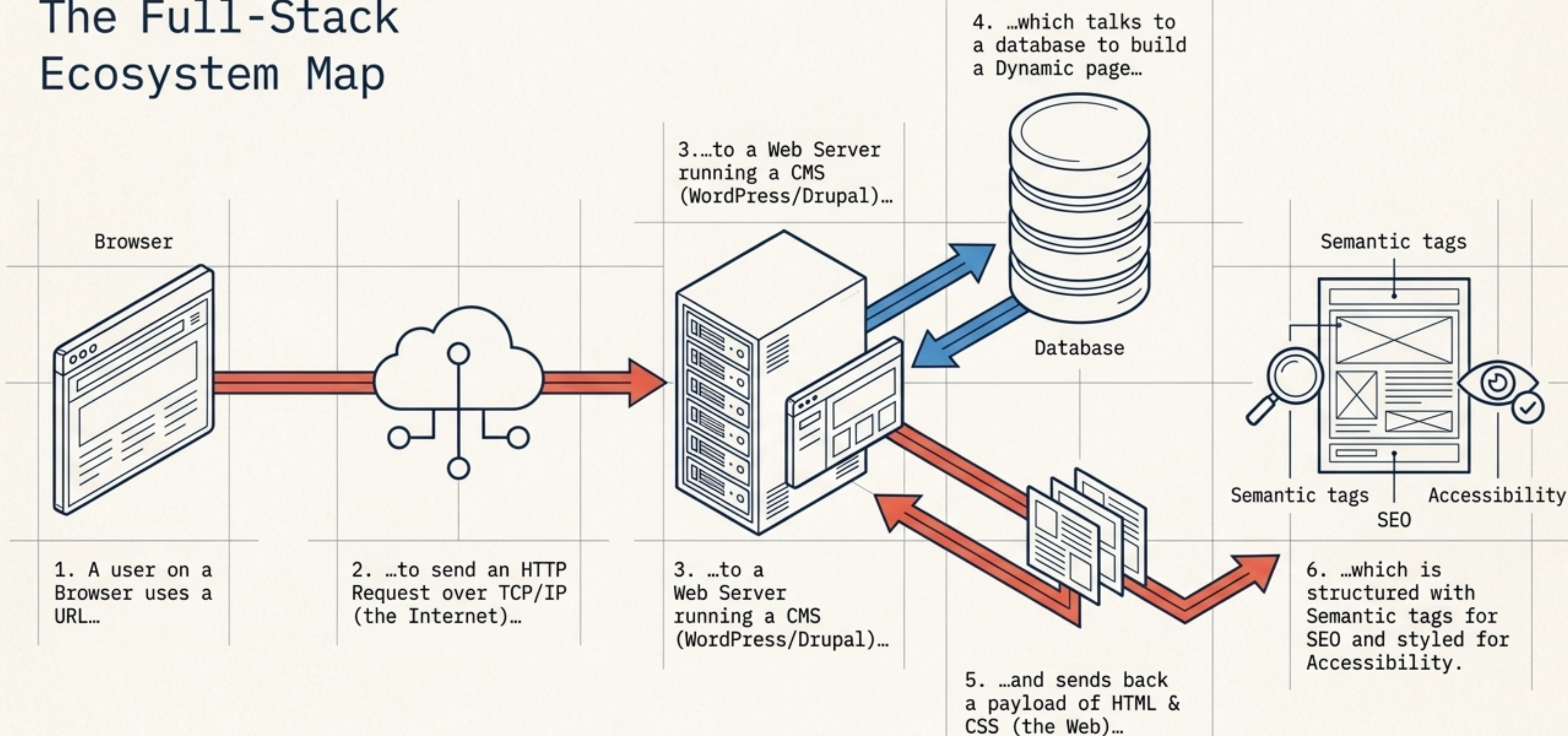


Joomla



Drupal

The Full-Stack Ecosystem Map



The Architect's Cheat Sheet



Network Layer: Internet = Global TCP/IP network; Web = Linked documents running on that network.



The Transaction: Browser (Client) requests via URL -> Web Server processes and responds.



The Address: URL dictates Protocol + Domain + Port + Path + Query + Fragment.



The Architecture: Static site = Fixed files; Dynamic site = Server-generated on request.



The Code: HTML = Structural semantics; CSS = Visual presentation; JS = Behavior.



The Protocol: Plan Purpose, Content, Layout, and Accessibility before writing code.